

```
1 C:\Users\thisa\AppData\Local\Microsoft\WindowsApps\
python3.13.exe C:\Users\thisa\Documents\
Software_Tech_1\Project\macro_project\src\console_app
.py
2
3 Macroinvertebrate Image Analysis System
4 =====
5 1. Show dataset summary (choose classes)
6 2. Generate EDA outputs
7 3. Train baseline classifier
8 4. Predict an image
9 5. Export dataset index to CSV
10 6. Show per-class image statistics
11 7. Exit
12 =====
13 Select an option: 1
14
15 --- Show Dataset Summary ---
16 Enter the path to your dataset folder: C:\Users\thisa
\Documents\Software_Tech_1\Project\macro_project\data
\raw\stream_macroinvertebrates
17
18 Found 17 classes:
19 1. Asellus sp
20 2. Baetidae sp
21 3. Elmia sp
22 4. Ephemerellidae
23 5. Erpobdella sp
24 6. Gammarus sp
25 7. Hydropsychidae sp
26 8. Leptophlebiidae sp
27 9. Leuctra sp
28 10. Limnias sp
29 11. Lymnea sp
30 12. Nemoura sp
31 13. Oligochaeta sp
32 14. Sericostomatidae sp
33 15. Sialis sp
34 16. Simuliidae sp
35 17. Sphaerium sp
36
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37   Enter the numbers of at least 3 classes to include
    , separated by commas.
38   Example: 1,3,5
39   Your selection: 1, 2, 3
40
41   Selected classes: Asellus sp, Baetidae sp, Elmis sp
42
43   === Dataset Summary ===
44   Total Images: 448
45   Total Classes: 3
46   Mean Width: 600.0
47   Mean Height: 370.5
48   Count Asellus Sp: 294
49   Count Baetidae Sp: 109
50   Count Elmis Sp: 45
51
52   Macroinvertebrate Image Analysis System
53   =====
54   1. Show dataset summary (choose classes)
55   2. Generate EDA outputs
56   3. Train baseline classifier
57   4. Predict an image
58   5. Export dataset index to CSV
59   6. Show per-class image statistics
60   7. Exit
61   =====
62   Select an option: 2
63
64   --- Generate EDA Outputs ---
65   EDA outputs saved to outputs/eda/
66
67   Macroinvertebrate Image Analysis System
68   =====
69   1. Show dataset summary (choose classes)
70   2. Generate EDA outputs
71   3. Train baseline classifier
72   4. Predict an image
73   5. Export dataset index to CSV
74   6. Show per-class image statistics
75   7. Exit
76   =====
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77   Select an option: 3
78
79   --- Train Baseline Classifier ---
80   ✓ Confusion matrix saved to C:\Users\thisa\Documents
    \Software_Tech_1\Project\macro_project\outputs\
    models\confusion_matrix.png
81   ✓ Classification report saved to C:\Users\thisa\
    Documents\Software_Tech_1\Project\macro_project\
    outputs\models\classification_report.png
82   ✓ Model saved to C:\Users\thisa\Documents\
    Software_Tech_1\Project\macro_project\outputs\models
    \macro_classifier.joblib
83   Accuracy: 0.8556
84
85   Classification Report:
86           precision    recall  f1-score
87   support
88   Asellus sp          0.83      0.98      0.90
89   Baetidae sp         1.00      0.64      0.78
90   Elmis sp            0.83      0.56      0.67
91
92   accuracy                                0.86
93   macro avg           0.89      0.72      0.78
94   weighted avg        0.87      0.86      0.85
95
96
97   Macroinvertebrate Image Analysis System
98   =====
99   1. Show dataset summary (choose classes)
100  2. Generate EDA outputs
101  3. Train baseline classifier
102  4. Predict an image
103  5. Export dataset index to CSV
104  6. Show per-class image statistics

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105     7. Exit
106     =====
107     Select an option: 4
108
109     --- Predict Image Class ---
110     Enter image path: C:\Users\thisa\Documents\
    Software_Tech_1\Project\macro_project\data\raw\
    stream_macroinvertebrates\Baetidae sp\CPH-Baetidae
    sp.-394-t.png
111     Predicted class: Baetidae sp
112     Confidence      : 97.50%
113
114     Macroinvertebrate Image Analysis System
115     =====
116     1. Show dataset summary (choose classes)
117     2. Generate EDA outputs
118     3. Train baseline classifier
119     4. Predict an image
120     5. Export dataset index to CSV
121     6. Show per-class image statistics
122     7. Exit
123     =====
124     Select an option: 5
125
126     --- Export Dataset Index to CSV ---
127     Enter the path to your dataset folder: C:\Users\
    thisa\Documents\Software_Tech_1\Project\
    macro_project\data\raw\stream_macroinvertebrates\
    Asellus sp
128     Index exported to: C:\Users\thisa\Documents\
    Software_Tech_1\Project\macro_project\outputs\
    reports\dataset_index.csv
129
130     Macroinvertebrate Image Analysis System
131     =====
132     1. Show dataset summary (choose classes)
133     2. Generate EDA outputs
134     3. Train baseline classifier
135     4. Predict an image
136     5. Export dataset index to CSV
137     6. Show per-class image statistics
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138     7. Exit
139     =====
140     Select an option: 6
141
142     --- Per-Class Image Statistics ---
143
144     Class                                Count    Mean W
145     Mean H    Min W    Max W    Min H    Max H
146     -----
147     Asellus sp                                294    600.0
148     350.2      600      600      338      450
149
150     Macroinvertebrate Image Analysis System
151     =====
152     1. Show dataset summary (choose classes)
153     2. Generate EDA outputs
154     3. Train baseline classifier
155     4. Predict an image
156     5. Export dataset index to CSV
157     6. Show per-class image statistics
158     7. Exit
159     =====
160     Select an option: 7
161
162     Exiting application. Goodbye.
163
164     Process finished with exit code 0
```